When Crypto Becomes Equity: A Risk-Adjusted Parity Optimized Model for the Post-2020 Comovement Shift

Daniel Felix Ahelegbey[3] Florian Gerth[4]
Oyakhilome Wallace Ibhagui[1,2]*

[1] Johns Hopkins University, USA

[2] BAUM TenPers Research Institute, USA

[3] University of Essex, UK

[4] Asian Institute of Management, Makati, Manila, Philippines

September 2025

Abstract

Recent research documents that major crypto assets no longer hedge equities; since 2020, their returns have visibly moved positively with stock markets. We propose a portable asset pricing model that potentially explains this shift. A representative investor allocates between a domestic equity index and a USD-denominated crypto asset and repatriates wealth each period if the domestic currency is not USD. The model gives rise to a risk-adjusted parity that links domestic equity returns to the FX-converted crypto leg (USD crypto return plus the bilateral FX change). The key mechanism is volatility convergence: as the volatility wedge between crypto and equities narrows, the variance bound implied by the parity tightens, pushing the equity–crypto loading strictly positive and toward unity. In this regime, crypto transmits equity-like risk rather than hedging it—consistent with the post-2020 breakdown of the "digital-gold" narrative noted by international authorities. The model suggests positive comovement after controlling for common shocks; stronger alignment when the wedge variance is small; and state-dependent deviations when the wedge briefly widens.

Keywords: Crypto assets, Equities, Risk-adjusted parity, Volatility wedge.
JEL codes: G12, F31, E44, G15.

*Corresponding author: `oibhagui@jhu.edu`

1 Introduction

For much of its early history, Bitcoin was heralded as "digital gold"—an asset expected to hedge equities or at least diversify conventional portfolios. Early empirical studies reported weak or state-contingent comovement with stocks and occasionally described hedge/safe-haven properties in specific samples or horizons (Dyhrberg, 2016; Bouri et al., 2017; Stensas et al., 2019; Urquhart, 2016). Since 2020, however, the stylized fact has flipped: multiple policy and academic sources document *persistently positive* comovement between major crypto assets and equities—especially in stress episodes—undermining the hedge narrative (Conlon and McGee, 2020; Adrian, Iyer, and Qureshi, 2022; International Monetary Fund, 2022; European Central Bank, 2025; Banco de España, 2022; Bank for International Settlements, 2025). This shift appears to be a stable feature of the post-2020 data-generating process rather than an anomaly.

This paper offers a theoretical explanation for that change. We propose a simple asset-pricing framework in which a representative investor allocates between a domestic equity index and a USD-denominated crypto asset and repatriates wealth each instant when the domestic currency is not USD or we set the exchange rate to 1 where the domestic currency is USD. The setup delivers risk-adjusted parity in local-currency real terms linking domestic equity returns to the FX-converted crypto leg (USD-crypto return plus the bilateral FX change). The key mechanism is volatility convergence: as the volatility "wedge" between crypto and equities narrows, the variance bound implied by parity tightens, forcing equity–crypto loading to be strictly positive and, in the limit, arbitrarily close to one. In such regimes, crypto transmits equity-like risk rather than hedging it.

Our contribution is conceptual and builds on observed regularities without invoking extra frictions or ad hoc preferences. First, parity explains why the post-2020 rise in equity–crypto correlation is not accidental but follows from a shrinking idiosyncratic wedge once portfolios and trading venues integrate. Second, the model predicts state dependence: when the wedge temporarily widens (e.g., liquidity freezes, or market segmentation), parity permits short-lived departures from tight comovement that subsequently compress as integration resumes. Third, the FX channel is explicit—crucial for economies where crypto payoffs are marked into local currency—so the same logic applies to both advanced and emerging markets.

Relation to the literature. Our analysis is motivated by (i) early, mixed evidence on hedge/safe-haven/diversifier roles (Dyhrberg, 2016; Bouri et al., 2017; Stensas et al., 2019); (ii) COVID-era findings that Bitcoin did *not* hedge equities and often amplified downside risk (Conlon and McGee, 2020; Chemkha et al., 2021); (iii) post-2020 policy and market surveillance showing sustained stock–crypto co-movement and growing traditional-finance linkages (Adrian, Iyer, and Qureshi, 2022; International Monetary Fund, 2022; European Central Bank, 2025; Banco de España, 2022; Bank for International Settlements, 2025); and (iv) evidence on crypto's distinctive factor structure and historically high (but evolving) volatility (Liu and Tsyvinski, 2018). We differ by providing a model that delivers positive, near-one loadings when the volatility wedge compresses, thereby rationalizing why the "digital-gold" narrative has broken down.

Roadmap. Section 3.1 develops the intuition and derives the risk-adjusted parity and the volatility-convergence result. Section **??** discusses extensions (heterogeneous investors, frictions, multiple tokens, stablecoins). We intentionally leave empirical validation to future work and use the observed post-2020 comovement only as motivation for the theory.

2 Stylized Fact for Intuition to Build Motivation

To build intuition for the idea in this paper, we track how the comovement between stock market returns and Bitcoin returns has evolved since the advent of Bitcoin trading, and how the *volatility wedge* between these two markets has changed over time. We also examine the wedge between the volatility of Bitcoin returns and equity returns, which helps assess whether the two assets remain distinct or increasingly behave as a single asset class.

As shown in Figure 1, Bitcoin initially exhibited little to no comovement with equities. In fact, the one-year rolling correlation of returns hovered near zero or negative values, supporting the early perception that Bitcoin could act as a hedge against equities. This period coincided with elevated volatility in the *returns wedge* (the difference between Bitcoin and equity returns) and in the *volatility wedge* (the difference in their rolling realized volatilities). In other words, Bitcoin's idiosyncratic risk profile dominated, creating a strong separation from the stock market and reinforcing its appeal as a diversification instrument.

Between 2016 and 2019, the volatility wedges began to decline markedly. As wedge volatility compressed, the correlation of Bitcoin with equities rose gradually, with several episodes of positive and statistically significant comovement. This suggests that Bitcoin's distinctiveness as an independent asset class started to erode as market forces aligned it more closely with equity performance.

Post-2019, the patterns become striking. The volatility wedges shrank to historically low levels, while the correlation between Bitcoin and equities surged—at times exceeding 0.55 on a one-year rolling basis in our sample. At higher frequency, monthly rolling correlations (not shown) occasionally surpass 0.90 over short intervals. This convergence implies that Bitcoin has lost much of its hedging or diversification property and now often amplifies equity market dynamics rather than offsetting them.

The COVID-19 shock in 2020 and the 2022 global monetary tightening provide clear stress tests. In both cases, correlations spiked unusually fast, highlighting Bitcoin's increasing tendency to transmit, rather than diversify, systemic shocks. These crisis dynamics reinforce a broader shift: Bitcoin behaves more like a high-beta equity factor than a diversifier.

Overall, the evidence points to a change in regime: Bitcoin's behavior is no longer governed by idiosyncratic drivers alone. As the volatility wedge declined, the "idiosyncratic gap" between Bitcoin and equities narrowed, and both assets increasingly loaded on common factors. This convergence is unlikely to reflect differences in pricing kernels, since both assets are priced under broadly similar discount factors in integrated markets. Instead, the compression of the volatility wedge appears to have eroded Bitcoin's uniqueness as an asset class, making its return dynamics far closer to equities.

From a market perspective, this stylized fact is crucial: the narrative of Bitcoin as "digital gold" or a hedge is increasingly untenable. Instead, Bitcoin's evolution suggests it now acts as a high-beta component of the equity market, at least relative to recent years.

For consistency, we use daily returns and one-year rolling windows to smooth short-term noise while retaining the visibility of the shifts discussed above. Although not reported here, alternative specifications using monthly windows produce the same qualitative conclusion.

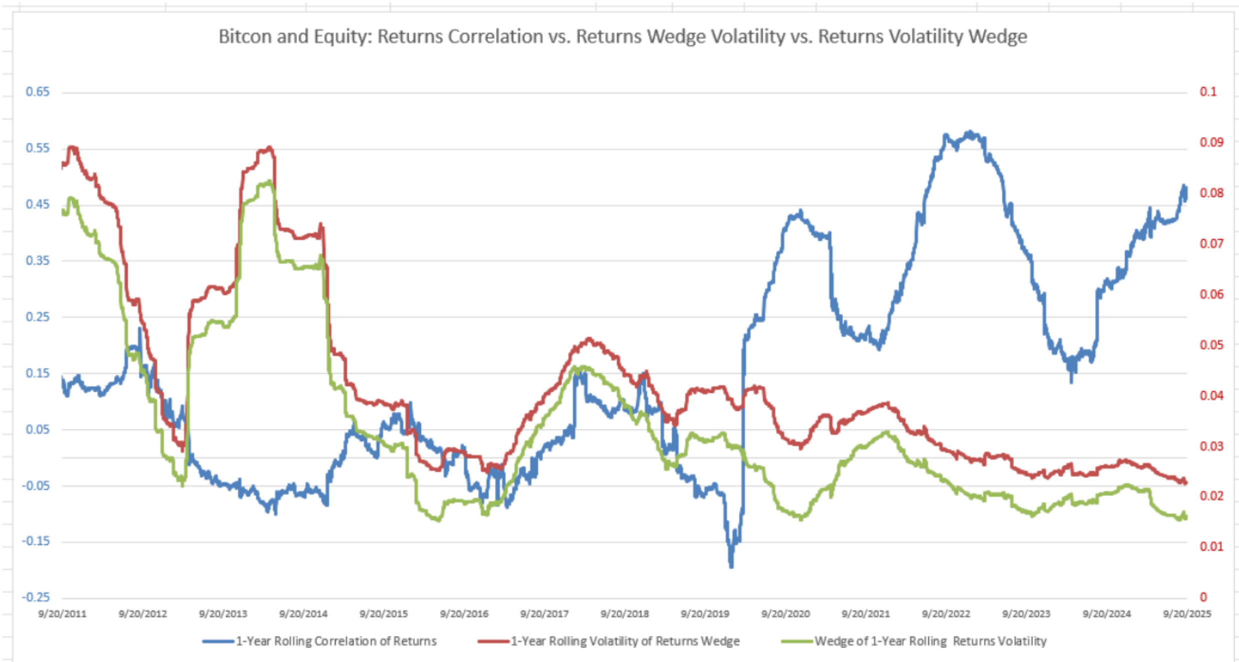

Figure 1: Bitcoin and Equity: 1-year rolling correlation of returns (left axis; blue) versus the volatility of the returns wedge (red) and the wedge of 1-year rolling returns volatility (green; right axis). The figure illustrates three phases: an early period of weak/negative comovement with high wedge volatility (2011–2016), a transition with compressing wedges and rising comovement (2016–2019), and a post-2019 convergence where correlations become persistently positive and high. Crisis episodes such as COVID-19 (2020) and the 2022 tightening cycle further reinforced Bitcoin's equity-like behavior.

3 A Model of Equity and Crypto Relationship

Motivated by the correlations, the volatility of returns, and the wedge evidence presented in the previous section, we now build a stylized model in which agents hold positions in a domestic equity and in Bitcoin (representative of a USD-denominated crypto asset). Because the investor's wealth is expressed in the domestic currency (LCU), whenever the unit of account differs from the U.S. dollar we must allow the exchange rate to enter the model explicitly. When flows are dollar-denominated or when the domestic currency *is* the dollar, we set the exchange rate to one without loss of generality.

Section 3.1 presents the model setup. Section 3.2 states the associated Bellman equation and solves the first-order conditions. Section 3.3 discusses the propositions implied by the model. Section 3.4 outlines potential extensions.

3.1 Model Setup

To concretize the idea in a tractable environment, consider a representative domestic investor who chooses consumption and portfolio holdings in discrete periods $t = 0, 1, 2, \ldots$. The investor can hold two risky assets: a domestic stock market claim and a global crypto asset (e.g., Bitcoin) that is priced in USD. At the end of each period, all wealth is marked to local currency (LCU). The investor receives exogenous labor income I_t in LCU.

Let C_t denote consumption and P_t^c the domestic price level (CPI, LCU). The gross inflation rate is

$$1 + \pi_{t+1} = \frac{P_{t+1}^c}{P_t^c}.$$

For the domestic stock market, let P_t^s be the stock price (LCU) and D_{t+1} the dividend (LCU) paid at t+1. The gross *nominal* and *real* returns are

$$1 + R_{t+1}^s = \frac{P_{t+1}^s + D_{t+1}}{P_t^s}, \qquad 1 + R_{t+1}^{s,\text{real}} = \frac{1 + R_{t+1}^s}{1 + \pi_{t+1}}.$$

For the cryptocurrency asset priced in USD, let $\Theta_{t+1}^\star$ be the USD return on the crypto asset (e.g., BTC) and define its log return $\theta_{t+1} \simeq \log(1 + \Theta_{t+1}^\star)$. Let S_t be the domestic exchange rate expressed as LCU per USD. The FX log return is

$$\Delta s_{t+1} = \log S_{t+1} - \log S_t,$$

where $\Delta s_{t+1} > 0$ means the dollar has appreciated against the LCU.

The investor chooses (i) the number of domestic shares n_t to carry from t to t+1 and (ii) the LCU amount A_t^k to allocate to the crypto leg at t. Investing A_t^k converts to a USD amount A_t^k/S_t at t, earns USD gross return $(1 + \Theta_{t+1}^\star)$ over $[t, t{+}1]$, and is repatriated at S_{t+1} into LCU at t+1.

3.2 Budget Constraint

The investor enters period t with positions (n_{t-1}, A_{t-1}^k). After receiving all inflows at t, she chooses (C_t, n_t, A_t^k) and carries only the two risky legs (equity and crypto) into t+1. Thus, in LCU,

Inflows at t:

$$\underbrace{(P_t^s + D_t)\, n_{t-1}}_{\text{stock sale + dividend}} + \underbrace{\frac{S_t}{S_{t-1}}\,(1 + \Theta_t^\star)\, A_{t-1}^k}_{\text{crypto proceeds in LCU after FX}} + \underbrace{I_t}_{\text{labor income}}.$$

Outflows at t:

$$\underbrace{P^c_t C_t}_{\text{consumption}} \quad + \quad \underbrace{P^s_t n_t}_{\text{new stock position}} \quad + \quad \underbrace{A^k_t}_{\text{new crypto allocation}} \quad .$$

Hence the flow budget is

$$P^c_t C_t + P^s_t n_t + A^k_t \;=\; (P^s_t + D_t) n_{t-1} + \frac{S_t}{S_{t-1}} (1 + \Theta^\star_t) A^k_{t-1} + I_t. \tag{1}$$

Define the LCU gross nominal and real returns on the crypto leg:

$$1 + R^k_{t+1} := \frac{S_{t+1}}{S_t}\,(1 + \Theta^\star_{t+1}), \qquad 1 + R^{k,\text{real}}_{t+1} = \frac{1 + R^k_{t+1}}{1 + \pi_{t+1}}. \tag{2}$$

In logs (first-order), $r^k_{t+1} \approx \theta_{t+1} + \Delta s_{t+1}$.

3.3 Investor's Discrete Time Optimization Problem

Following standard arguments, we assume preferences are time-separable with discount factor $0 < \beta < 1$ and strictly increasing, concave $u(\cdot)$. The investor chooses $\{C_t, n_t, A^k_t\}_{t\geq 0}$ to solve the optimization problem in discrete time space

$$\max\ \mathbb{E}_0 \sum_{t=0}^{\infty} \gamma^t u(C_t) \quad \text{s.t.} \quad (1), \quad n_t \geq 0,\ A^k_t \geq 0. \tag{3}$$

Define cash-on-hand (LCU) at time t:

$$M_t := (P^s_t + D_t)\, n_{t-1} \;+\; \frac{S_t}{S_{t-1}} (1 + \Theta^\star_t)\, A^k_{t-1} \;+\; I_t.$$

The investor chooses (C_t, n_t, A^k_t) to solve the Bellman problem

$$V(M_t) \;=\; \max_{C_t, n_t, A^k_t} \Big\{ u(C_t) \;+\; \gamma\, \mathbb{E}_t \big[V(M_{t+1}) \big] \Big\}$$

subject to

$$P^c_t C_t \;+\; P^s_t n_t \;+\; A^k_t \;=\; M_t, \qquad M_{t+1} \;=\; (P^s_{t+1} + D_{t+1})\, n_t \;+\; \frac{S_{t+1}}{S_t}\,(1 + \Theta^\star_{t+1})\, A^k_t \;+\; I_{t+1}.$$

Let $V_M(\cdot)$ denote the derivative of V with respect to cash-on-hand. The standard first-order conditions lead to Euler equations in real LCU. Let the real SDF be

$$m_{t+1} := \gamma \, \frac{u'(C_{t+1})}{u'(C_t)} \frac{1}{1+\pi_{t+1}}.$$

Then

$$\mathbb{E}_t\big[m_{t+1}\,(1+R^s_{t+1})\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R^k_{t+1})\big] = 1,$$

and, equivalently in real terms,

$$\mathbb{E}_t\big[m_{t+1}\,(1+R^{s,\text{real}}_{t+1})\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R^{k,\text{real}}_{t+1})\big] = 1.$$

Subtracting, we obtain the *risk-adjusted parity*:

$$\boxed{\mathbb{E}_t\Big[m_{t+1}\Big((1+R^s_{t+1}) - (1+R^k_{t+1})\Big)\Big] = 0} \iff \mathbb{E}_t\big[m_{t+1}\,(R^{s,\text{real}}_{t+1} - R^{k,\text{real}}_{t+1})\big] = 0.$$

Let $w_{t+1} := R^{s,\text{real}}_{t+1} - R^{k,\text{real}}_{t+1}$. From the parity condition we have

$$E_t[w_{t+1}] = -\frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]}. \tag{2}$$

Decompose w_{t+1} around its conditional mean:

$$w_{t+1} = E_t[w_{t+1}] + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{3}$$

$$\varepsilon_{t+1} := w_{t+1} - E_t[w_{t+1}]. \tag{3'}$$

Since $R^{s,\text{real}}_{t+1} = R^{k,\text{real}}_{t+1} + w_{t+1}$, it follows that

$$R^{s,\text{real}}_{t+1} = R^{k,\text{real}}_{t+1} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{4}$$

Corollary Under the model condition, projecting $R^{s,\text{real}}_{t+1}$ on $R^{k,\text{real}}_{t+1}$ yields

$$R^{s,\text{real}}_{t+1} = \alpha + \beta\, R^{k,\text{real}}_{t+1} + u_{t+1}, \qquad E_t[u_{t+1}] = 0,$$

where α captures the average mean adjustment, β measures the strength of comovement, and u_{t+1} is the mean-zero wedge innovation. The model predicts $\beta > 0$ and $\beta \to 1$ as the idiosyncratic wedge variance vanishes.

The structural parity condition implies

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0,$$

so the conditional mean of the wedge is determined by the interaction of the pricing kernel m_{t+1} with the wedge w_{t+1}. In the empirical regression

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1},$$

this adjustment cannot be observed directly. Its average effect is absorbed into the intercept α, while any time-varying deviations not captured by α appear in the residual u_{t+1}. Thus α and u_{t+1} together proxy for the SDF–wedge term in empirical implementation.

We now generate the main testable proposition linking the domestic equity and crypto assets. We now state two testable propositions that link the domestic equity leg and the FX-converted crypto leg implied by the first order condition.

3.4 Main Proposition

In the short run, after controlling for common shocks,

$$r_{t+1}^{s,\text{real}} = \alpha + \beta\, r_{t+1}^{k,\text{real}} + u_{t+1}.$$

Let ε_{t+1}^{s} and ε_{t+1}^{k} be the idiosyncratic components of $r_{t+1}^{s,\text{real}}$ and $r_{t+1}^{k,\text{real}}$ after removing common shocks, and define the idiosyncratic parity wedge $e_{t+1} := \varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k}$. Then

$$\beta = \frac{\text{Cov}(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{\text{Var}(\varepsilon_{t+1}^{k})} = 1 + \frac{\text{Cov}(e_{t+1}, \varepsilon_{t+1}^{k})}{\text{Var}(\varepsilon_{t+1}^{k})},$$

and by Cauchy–Schwarz,

$$1 - \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^{k})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{\text{Var}(e_{t+1})}{\text{Var}(\varepsilon_{t+1}^{k})}}.$$

In particular, if $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^{k}) < 1$, then $\beta \in (0, 2)$ and is therefore positive; if $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^{k}) = 1$, then $\beta \in [0, 2]$; and if $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^{k}) > 1$, then $\beta \in \mathbb{R}$, implying that β may become negative when the wedge variance is sufficiently large. Moreover, as $\text{Var}(e_{t+1})/\text{Var}(\varepsilon_{t+1}^{k}) \to 0$, we have $\beta \to 1$, as required.

Economic intuition. When the idiosyncratic wedge e_{t+1} is small relative to the volatility of the FX-converted crypto leg, the regression slope is forced to lie strictly above zero and concentrate near one. The inequality quantifies this: the tighter the residual wedge, the tighter β is around one. In other words, crypto and domestic equities are positively linked and comove in the short run, rather than offsetting each other. The smaller is the variance of their idiosyncratic wedge, the stronger is the relationship between crypto and equities. This also suggests that periods where equities and crypto returns have offset each other or exhibited very weak links are periods characterized by large wedge variance. Previously, Bitcoin and equities appeared unrelated because the wedge variance was large: thin liquidity, fragmented markets, and regulatory frictions meant crypto returns deviated significantly from equity-linked parity, widening the bounds around β and making estimated correlations indistinguishable from zero. More recently, large institutional, individual, or even sovereign adoption, portfolio integration, and ETF/futures access compressed wedge variance, with the relationship concentrating at higher values and explaining why correlations jumped and stayed persistently positive. Thus, both crypto and equities load on similar factor, and the smaller the wedge variance, the stronger both assets comove and the closer the slope is to one. Our model shows how crypto can shift from offsetting equity risk to transmitting it—a move from the digital gold narrative to equity-like behavior.

3.5 Extensions

The baseline model is intentionally stylized, designed to isolate the core equity–crypto relationship through the FX channel. Nevertheless, several natural extensions could be considered. First, one could introduce heterogeneity in investor types, distinguishing between financially integrated households and liquidity-constrained agents. This would allow wedge variance to depend on market participation and access, potentially explaining cross-country heterogeneity. Second, relaxing the assumption of frictionless markets to incorporate capital controls, liquidity freezes, or transaction costs would make explicit the mechanisms by which wedge variance spikes during stress. Third, the model could be extended to multiple crypto assets (e.g., Bitcoin and Ether), or to stablecoins alongside volatile tokens, to capture a richer menu of USD-linked instruments. Importantly, none of these extensions would overturn the main result of the Proposition: as long as both legs load on the same SDF, the short-run slope remains strictly positive and close to unity whenever the idiosyncratic wedge variance is small.

4 Structural Simulation

To visualize the theoretical predictions derived in Section 3.1, we simulate a synthetic data-generating process (DGP) that operationalizes the relationship between the wedge variance and the equity–crypto slope β. The goal is not calibration to actual data, but rather to demonstrate numerically how β converges toward unity as the idiosyncratic parity wedge vanishes.

4.1 Design of the Data-Generating Process

The DGP captures three elements of the theoretical setup: (i) a common systematic factor that both assets load on equally, (ii) independent idiosyncratic noise in the crypto leg, and (iii) a wedge process representing departures from the risk-adjusted parity.

Common factor. Both the domestic equity and crypto returns share a common component F_{t+1}, interpreted as a broad market or macro-financial factor. We assume

$$F_{t+1} \sim \mathcal{N}(0, \sigma_F^2),$$

and both assets have identical loadings b on this factor. This symmetry ensures that, absent idiosyncratic disturbances, the two returns would move one-for-one.

Idiosyncratic crypto noise. The crypto return includes an additional idiosyncratic disturbance ε_{t+1}^k capturing coin-specific or microstructure risk, drawn from

$$\varepsilon_{t+1}^k \sim \mathcal{N}(0, \sigma_k^2),$$

independent of F_{t+1}.

Parity wedge process. Departures from the risk-adjusted parity are modeled by the idiosyncratic wedge e_{t+1}, defined as the residual difference between the equity and crypto legs after removing common shocks. The wedge follows a two-state Markov process capturing regimes of financial integration:

$$e_{t+1} \,|\, s_t \sim \mathcal{N}(0, \sigma_{e,s_t}^2), \qquad s_t \in \{\text{tight}, \text{loose}\},$$

where $\sigma_{e,\text{tight}} < \sigma_{e,\text{loose}}$. Intuitively, when markets are tightly integrated, the wedge variance is small; when fragmented or segmented, the wedge variance is large. We also allow state-dependent correlation $\rho_s := \mathrm{Corr}(e_{t+1}, \varepsilon^k_{t+1} \mid s_t)$ to represent whether the deviation co-moves with or offsets the crypto shock within each regime.

4.2 Return Equations

Real returns for the crypto and equity legs are generated as:

$$\begin{aligned} r^{k,\text{real}}_{t+1} &= bF_{t+1} + \varepsilon^k_{t+1}, \\ r^{s,\text{real}}_{t+1} &= bF_{t+1} + \varepsilon^k_{t+1} + e_{t+1}. \end{aligned}$$

Hence, e_{t+1} directly measures the violation of the risk-adjusted parity condition. When $e_{t+1} = 0$, the model implies $\beta = 1$; when e_{t+1} is volatile, β departs from unity in proportion to the wedge variance.

4.3 Estimation and Rolling Computation

We generate 2,000 synthetic return pairs $(r^{s,\text{real}}_{t+1}, r^{k,\text{real}}_{t+1})$ with regime switching between tight and loose states. To estimate the short-run equity–crypto linkage, we compute rolling regressions of $r^{s,\text{real}}_{t+1}$ on $r^{k,\text{real}}_{t+1}$ while partialling out the common factor F_{t+1}. Each window contains 250 observations, approximating one trading year in daily data.

The estimated slope coefficients $\hat{\beta}_t$ are plotted alongside the corresponding rolling wedge-variance ratio

$$\phi_t = \frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})},$$

computed within the same window. The time-series visualization in Figure 2 shows that $\hat{\beta}_t$ rises toward unity when ϕ_t is small (tight-integration regime) and declines when ϕ_t is large (loose-integration regime), reproducing the theoretical bounds.

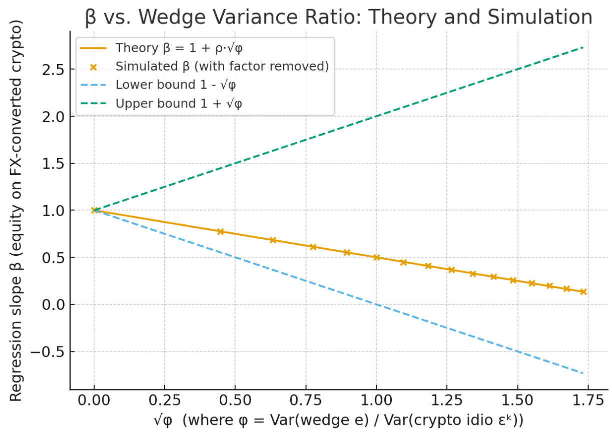

Figure 2: Cross-sectional check: simulated β vs. $\sqrt{\phi}$ alongside the theoretical line $1 + \rho\sqrt{\phi}$ and the bounds $[1 - \sqrt{\phi},\, 1 + \sqrt{\phi}]$.

4.4 Cross-Sectional Verification

To confirm the theoretical inequality, we run a grid simulation varying the wedge-to-crypto variance ratio $\phi = \sigma_e^2/\sigma_k^2$ and the correlation ρ. For each (ϕ, ρ) pair, we estimate the average β from simulated data and compare it with the theoretical expression $1 + \rho\sqrt{\phi}$. Figure 3 demonstrates close alignment between simulated and theoretical slopes, with all estimates contained within the analytical bounds in inequality (4):

$$1 - \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}}. \tag{4}$$

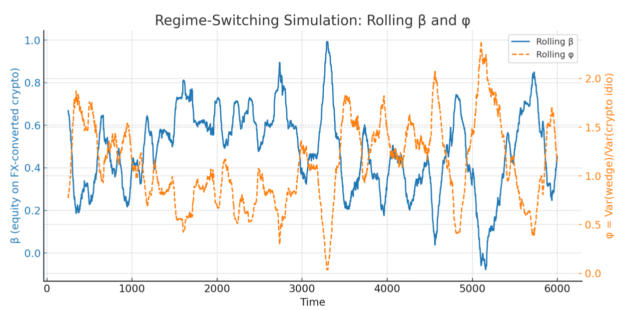

Figure 3: Rolling $\hat{\beta}_t$ (equity on FX-converted crypto, common factor partialled out) and rolling wedge-variance ratio ϕ_t computed over 250-observation windows.

The simulation verifies the central message of the model. In the tight-integration regime, where market segmentation is minimal and parity nearly holds, β concentrates around one: crypto transmits equity-like risk. In the loose-integration regime, where parity violations are large, β declines or can turn slightly negative. The dynamic relationship between β and the wedge variance ϕ thus provides a quantitative representation of how integration strength determines the sign and magnitude of crypto–equity comovement.

5 Market and Policy Implications

Our parity framework yields concrete implications for how markets should *use* crypto exposures and how authorities should *oversee* them once crypto behaves as an equity-like risk when the volatility wedge is small.

5.1 Portfolio construction and risk management

Do not treat crypto as a hedge. When the volatility wedge is compressed, our model forces a strictly positive short-run loading of equities on the FX-converted crypto leg. Practically, BTC (and similar tokens) should be budgeted as a high-beta equity extension rather than a diversifier. This is consistent with the post-2020 rise in stock–crypto comovement documented by international policy work (IMF, 2022; ECB, 2025; BIS, 2025).

Risk budgets and overlays. Risk teams should cap crypto exposure inside the *equity* bucket, not the *alternatives/hedge* bucket; hedges for crypto drawdowns should be equity-style (index puts, volatility overlays), not gold/UST by default. In our model, hedge effectiveness improves when one hedges the common component directly and recognizes that the idiosyncratic parity wedge is the only piece not spanned by equities.

State dependence & triggers. Because comovement tightens endogenously when the wedge variance shrinks, institutions should use regime-switching risk limits: (i) a *tight-comovement regime* (small wedge variance) where crypto risk is consolidated with equity risk; (ii) a *loose-comovement regime* (large wedge variance) during market segmentation, liquidity freezes, or capital-control episodes, where crypto can temporarily decouple. Empirically, public surveillance has noted both the *rise* in average comovement and the persistence of episodic decouplings around market frictions and liquidity stress (IMF, 2022; ECB, 2025).

Stress testing. Scenario design should include (a) high-volatility, tight-comovement shocks where equity and crypto fall together; and (b) segmentation shocks where the wedge briefly widens and re-compresses. The wedge process is the right "knob" in scenarios; it directly maps into the slope that binds parity in our model.

5.2 Access channels and market plumbing

ETFs deepen integration. Spot BTC ETFs approved in the U.S. (Jan 10, 2024) create a low-friction bridge from TradFi portfolios into crypto, reinforcing the conditions under which our parity bites (smaller wedge, tighter comovement). Portfolio access via ETFs makes crypto exposures operationally indistinguishable from equity beta, accelerating the migration from "hedge" to "transmission channel" (SEC statements; U.S. CRS note; contemporaneous market summaries).

FX pass-through in EMs. Because our leg is USD-denominated and repatriated to LCU, FX changes transmit directly into local-currency payoffs. In depreciation episodes, the local equity and the FX-converted crypto leg can *both* amplify drawdowns in real terms; this is a key reason EM policymakers should watch on/off-ramps and *USD* funding conditions even for "domestic" crypto activity (IMF work on cross-border flows offers related measurement motivation).

5.3 Prudential and macroprudential oversight

Bank exposures and capital. Our model implies that when the wedge is small, crypto behaves like high-beta equity risk. The Basel Committee's crypto standard—with conservative capital for unbacked crypto and tighter risk controls—is directionally consistent with that interpretation; 2024 amendments clarified the prudential treatment and monitoring expectations ahead of implementation (BCBS, 2024; BIS communications).

Systemic surveillance. Authorities increasingly flag stronger stock–crypto comovement and deeper TradFi links as a stability issue (IMF policy notes, ECB FSRs, BIS reports). Our parity offers a structural lens: a shrinking wedge explains persistent correlation without invoking new preference shocks; it also tells supervisors *which knob to watch*—integration/liquidity frictions that move the wedge. Macroprudential tools should therefore target on/off-ramp liquidity, prime-broker intermediation, and leveraged ETF/activity channels that compress wedges further during risk-on phases (IMF, 2022; ECB, 2025; BIS, 2025).

Stablecoins and spillovers. Where stablecoins intermediate flows between banks, funds, and exchanges, reserve-asset composition pulls crypto closer to traditional markets; this can narrow wedges mechanically and propagate shocks across balance sheets (IMF, 2025; BIS commentary). Supervisors should align reserve quality, disclosure, and run-risk safeguards with money-market fund standards.

5.4 Policy design for emerging markets

Capital-flow and FX considerations. If residents use USD-crypto as a de facto external asset, FX policy and capital-flow management interact with parity: USD strength and on-ramp frictions jointly set wedge dynamics and comovement. The clean implication is *not* to ban crypto, but to (i) ensure visibility on cross-border channels, (ii) level-up reserve/balance-sheet statistics to include crypto-linked flows, and (iii) supervise the fiat gateways where wedge compression is most forceful (policy sources emphasize data gaps and cross-market links).

5.5 What the model adds, in practice

- **Classification.** Treat major crypto as *equity-adjacent risk* when wedge variance is small; reclassify to *niche/uncorrelated* only in segmentation regimes.
- **Measurement.** Monitor a *wedge variance* metric (e.g., residual variance after removing common equity/FX factors). It is the sufficient statistic that pins the equity–crypto slope in our model.
- **Scenarios.** Stress tests should shock the wedge directly (tighten/loosen) alongside equity/FX shocks.
- **Governance.** Risk limits, collateral haircuts, and margin models should step up when the wedge compresses; that is exactly when comovement—and therefore systemic beta—is *strongest*.

Bottom line. Our parity makes a sharp, testable prediction that lines up with post-2020 experience: once integration compresses the idiosyncratic wedge, crypto ceases to hedge equities and instead *transmits* equity-like risk. Portfolio policy should therefore budget crypto like equity, and supervisory policy should focus less on whether crypto is "uncorrelated" and more on the plumbing that compresses or loosens the wedge (access via ETFs; stablecoin reserves; on/off-ramp liquidity; leverage conditions). The policy record already points in that direction (IMF/ECB/BIS, 2022–2025; SEC 2024; BCBS 2024).

6 Conclusion

This paper develops a parsimonious, optimization based discrete-time framework that explains why major crypto assets have ceased to hedge equities and instead comove positively with them. The core result is a risk-adjusted parity between a domestic equity index and a USD-denominated crypto leg once payoffs are marked to local currency: when portfolio and market integration compress the idiosyncratic wedge between the two legs, the model implies a strictly positive short-run loading of equities on the FX-converted crypto return—and, in the limit of a vanishing wedge, that loading tends to unity. In this regime, crypto transmits equity-like risk rather than offsetting it, offering a structural rationale for the post-2020 breakdown of the "digital gold" narrative documented by policy and market surveillance (IMF, 2022; BIS, 2025; ECB, 2025).

Conceptually, three advances follow. First, the theory unifies disparate empirical regularities without adding frictions or ad hoc preferences: parity alone explains why correlations rise as volatility differentials narrow. Second, the mechanism is state dependent: temporary market segmentation, liquidity freezes, capital controls, or FX bottlenecks widen the wedge and permit brief decouplings; as integration resumes, the variance bound tightens and co-movement re-converges. Third, the FX channel is explicit, so the same logic applies in both advanced and emerging markets where crypto payoffs are repatriated to local currency.

These insights carry practical implications. Portfolio policy should budget major crypto exposures as equity-adjacent beta whenever wedge variance is small; risk overlays and stress tests should shock the wedge directly, alongside equity and FX factors. On the supervisory side, the lens is consistent with tighter prudential treatment for unbacked crypto and heightened systemic surveillance as TradFi linkages deepen (BCBS, 2024; BIS, 2025). Market-access innovations (such as spot Bitcoin ETFs) compress frictions and can accelerate the transition from "hedge" to "transmission channel" (SEC, 2024).

The model is intentionally minimal. That minimalism clarifies where future work can add richness without diluting the central message: heterogeneous investors and intermediation, explicit trading frictions and on/off-ramp bottlenecks, multiple tokens (including stablecoins), and endogenous wedge dynamics. Each extension preserves the core parity logic while delivering testable predictions—most directly, that the equity–crypto slope increases toward one as wedge variance falls. In short, once integration compresses the idiosyncratic wedge, crypto's portfolio and policy role shifts from diversification to amplification. –

References

[1] Adrian, T., Iyer, R., & Qureshi, M. (2022). Crypto Prices Move More in Sync With Stocks, Posing New Risks. *IMF Blog*, January 11, 2022.

[2] Banco de España. (2022). *Financial Stability Report* (Spring 2022). (Multiple articles

discuss crypto–financial linkages and market co-movements.)

[3] Basel Committee on Banking Supervision (BCBS). (2022). *Prudential treatment of cryptoasset exposures*. Basel, December 2022. (Final standard.)

[4] Bank for International Settlements (BIS). (2022). *BIS Annual Economic Report 2022: Chapter III – The future monetary system*. Basel: BIS.

[5] Bank for International Settlements (BIS). (2023). *Crypto shocks and retail losses*. BIS Bulletin No. 66.

[6] Bouri, E., Molnár, P., Azzi, G., Roubaud, D., & Hagfors, L. I. (2017). On the hedge and safe haven properties of Bitcoin: Is it really more than a diversifier? *Finance Research Letters*, 20, 192–198.

[7] Chemkha, R., Benkraiem, R., & others (2021). Safe-haven properties of cryptocurrencies during COVID-19: A careful re-examination. *(Journal article—update with your exact version used).*

[8] Conlon, T., & McGee, R. (2020). Safe haven or risky hazard? Bitcoin during the COVID-19 bear market. *Finance Research Letters*, 35, 101607.

[9] Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar—A GARCH volatility analysis. *Finance Research Letters*, 16, 85–92.

[10] European Central Bank (ECB). (2023). *Financial Stability Review* (May 2023). (Contains essays on crypto-asset developments and financial stability implications.)

[11] International Monetary Fund (IMF). (2022). *Global Financial Stability Report* (October 2022). (Discusses rising stock–crypto co-movement and financial-stability considerations.)

[12] Liu, Y., & Tsyvinski, A. (2021). Risks and Returns of Cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.

[13] U.S. Securities and Exchange Commission (SEC). (2024). Order Granting Approval of Proposed Rule Changes to List and Trade Shares of Spot Bitcoin Exchange-Traded Products. (Approved January 10, 2024.)

[14] Stensås, A., Nygaard, M. F., Kyaw, K. S., & Klovland, J. T. (2019). Can Bitcoin be a diversifier, hedge or safe haven for equity investors? *International Review of Financial Analysis*, 63, 322–330.

[15] Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.

Appendix A. Proofs and Theoretical Results

This appendix provides proofs for the model implications in Section 3.1 and the formal justification for main Proposition Throughout, expectations are conditional on $\mathcal{F}_t$ and all returns are *gross* unless otherwise noted. We assume interior solutions ($n_t > 0$, $A_t^k > 0$) and that $\{C_t\}$ is strictly positive and adapted.

A.1. Parity Condition

Let m_{t+1} denote the real LCU stochastic discount factor

$$m_{t+1} := \gamma \, \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t^c}{P_{t+1}^c} = \gamma \, \frac{u'(C_{t+1})}{u'(C_t)} \frac{1}{1 + \pi_{t+1}}.$$

Then any tradable payoff $1 + R_{t+1}$ quoted in nominal LCU satisfies $E_t[m_{t+1}(1 + R_{t+1})] = 1$ and equivalently in real terms $E_t[m_{t+1}(1 + R_{t+1}^{\text{real}})] = 1$.

Proof. From the Bellman problem in §3.3, the Lagrange multiplier on the period-t budget constraint is $\lambda_t = u'(C_t)/P_t^c$. FOC for n_t and A_t^k give, respectively,

$$\lambda_t P_t^s = \beta \, E_t[\lambda_{t+1}(P_{t+1}^s + D_{t+1})], \qquad \lambda_t = \beta \, E_t\left[\lambda_{t+1} \frac{S_{t+1}}{S_t}(1 + \Theta_{t+1}^{\star})\right].$$

Divide by λ_t, substitute $\lambda_{t+1}/\lambda_t = (u'(C_{t+1})/u'(C_t))(P_t^c/P_{t+1}^c)$, and rearrange to obtain $E_t[m_{t+1}(1 + R_{t+1}^s)] = 1$ and $E_t[m_{t+1}(1 + R_{t+1}^k)] = 1$. Dividing each return by $1 + \pi_{t+1} = P_{t+1}^c/P_t^c$ gives the real versions. □

For the domestic equity and FX-converted crypto legs in real LCU,

$$E_t\big[m_{t+1}\big((1 + R_{t+1}^{s,\text{real}}) - (1 + R_{t+1}^{k,\text{real}})\big)\big] = 0.$$

Equivalently, letting $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$,

$$E_t[w_{t+1}] = -\frac{Cov_t\big(m_{t+1}, \, w_{t+1}\big)}{E_t[m_{t+1}]} \, . \tag{5}$$

A.2. Proof of Main Proposition

Let the factor decomposition capture the shocks common to the two legs:

$$r_{t+1}^{s,\text{real}} = \mu_s + b' F_{t+1} + \varepsilon_{t+1}^{s},$$
$$r_{t+1}^{k,\text{real}} = \mu_k + b' F_{t+1} + \varepsilon_{t+1}^{k},$$

where F_{t+1} stacks common risk shocks (including FX and global conditions) and the idiosyncratic components $\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k}$ are mean zero and orthogonal to F_{t+1}. Time fixed effects λ_t span the space of F_{t+1} (up to estimation noise), so the Frisch–Waugh–Lovell theorem implies that the coefficient β in the regression of $r_{t+1}^{s,\text{real}}$ on $r_{t+1}^{k,\text{real}}$ *controlling for* λ_t equals the slope in the auxiliary regression of the residuals:

$$\underbrace{(r_{t+1}^{s,\text{real}} - \mu_s - b' F_{t+1})}_{\varepsilon_{t+1}^{s}} \;=\; \beta\; \underbrace{(r_{t+1}^{k,\text{real}} - \mu_k - b' F_{t+1})}_{\varepsilon_{t+1}^{k}} + u_{t+1}.$$

Hence,

$$\beta = \frac{Cov(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} = 1 - \frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{2Var(\varepsilon_{t+1}^{k})} \;+\; \frac{Var(\varepsilon_{t+1}^{s}) - Var(\varepsilon_{t+1}^{k})}{2Var(\varepsilon_{t+1}^{k})}. \tag{6}$$

$$\beta = \frac{Cov(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} = 1 + \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}. \tag{7}$$

By Cauchy–Schwarz,

$$\left|Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})\right| \;\le\; \sqrt{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})\, Var(\varepsilon_{t+1}^{k})}.$$

Equivalently, the two-sided bound is

$$-\sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}} \;\le\; \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} \;\le\; \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}.$$

Thus,

$$1 - \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}} \;\le\; 1 + \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} \;\le\; 1 + \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}.$$

Substituting into

$$\beta = 1 + \frac{Cov(\varepsilon^s_{t+1} - \varepsilon^k_{t+1},\, \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})},$$

we obtain the bound

$$1 - \sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}.$$

In the absent large state-dependent risk adjustments, we have that $\varepsilon^s_{t+1} \approx \varepsilon^k_{t+1}$ and $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})$ is small. Therefore, substitute to obtain

$$\beta \;=\; 1 \;+\; O\!\left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right),$$

so $\beta \approx> 1$ and, in particular, $\beta > 0$ whenever $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1}) < Var(\varepsilon^k_{t+1})$.

or

$$\beta \;=\; 1 \;-\; O\!\left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right),$$

so $\beta \approx> 1$ and, in particular, $\beta > 0$ whenever $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1}) < Var(\varepsilon^k_{t+1})$.

where

$$O\!\left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right) \;\to\; 0 \quad \text{as} \quad \left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right), \;\to\; 0,$$

because $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1}) \to 0$.

Therefore, by the variance bound

$$1 - \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}},$$

it follows that $\beta > 0$ whenever the idiosyncratic discrepancy between the two assets is small relative to the variation in the crypto leg, and β becomes arbitrarily close to unity as that discrepancy converges to zero. Conversely, when the wedge variance dominates—that is, when $\mathrm{Var}(e_{t+1})/\mathrm{Var}(\varepsilon^k_{t+1}) > 1$—the bound permits β to enter negative territory, reflecting episodes in which idiosyncratic shocks overpower the systematic parity linkage. Hence, the sign and magnitude of β provide a direct empirical map from wedge variance to the strength and direction of equity–crypto comovement: tight integration forces $\beta \approx 1$, while loose integration allows β to collapse or even invert.

In essence, the slope β becomes a sufficient statistic for market integration: its convergence toward one marks the absorption of crypto into the equity risk complex, while its dispersion or sign reversal signals the re-emergence of segmentation and friction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declarations

Funding

This research received no external funding.

Data availability

The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics approval

Not applicable.

Consent to participate

Not applicable.

Consent for publication

All authors consent to the publication of this manuscript.

Competing interests

The authors declare that they have no competing interests.